

Program : Diploma in Tool & Die Engineering	
Course Code : 4118	Course Title: Moulds & Dies–Heat Treatment Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To familiarize students with the Injection/compression/transfer Moulding principles and operations in Tool & Die making.
- To provide skills on various precision measuring instruments for taking dimensions.
- To impart practical knowledge in heat treatment and skill development on its industrial application.
- To give knowledge on industrial assembly & maintenance of various kind injection moulds.

Course Prerequisites:

Topic	Course code	Course name	Semester
Design concepts of Moulds& Dies		Moulds& Dies Technology	3
Limits, Fits, G&D Tolerance, Surface Finish symbols and values		Tool Engineering drawing	3
Heat treatment processes & various tool steels		Material science & Metrology	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Produce the mould components of hand injection Mould/compression mould/transfer mould	15	Applying
CO2	Practice the process of core & cavity blue matching.	6	Applying

CO3	Demonstrate the Industrial mould assembly & maintenance.	12	Applying
CO4	Apply various heat treatment processes on different types of tool steels.	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						3
CO2	3						
CO3	3						
CO4	3						3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Produce the mold components of hand injection Mold/compression mold/transfer mould		
M1.01	Produce various Mould components – Cavity plate, core plate, Ejector Assembly, Guide Pins & Pillars etc.	15	Applying
CO2	Practice the process of core & cavity blue matching.		
M2.01	Demonstrate blue matching of cavity and core inserts by means of a thin and semitransparent layer of Prussian blue.	2	Applying
M2.02	Demonstrate the micro finishing operations Lapping, Honing, superfinishing etc.	2	Applying
M2.03	Employ the acceptance criteria and do the necessary rectifications.	2	Applying
	Lab Exam – 1	1.5	
CO3	Demonstrate the Industrial mould assembly & maintenance.		
M3.01	Describe the working of Hand injection/Injection moulding machines	3	Understanding
M3.02	Apply a Product Trial run on the moulding machine and check for any flash.	3	Applying

M3.03	Employ the acceptance criteria and do the necessary rectifications.	1	Applying
M3.04	Practice on assembly & maintenance of various types industrial moulds – single cavity moulds, multi cavity moulds, two plate moulds.	5	Applying
CO4	Apply various heat treatment processes on different types of tool steels.		
M 4.01	Describe different heat treatment processes, its purposes and the obtained properties.	3	Understanding
M 4.02	Practice different heat treatment processes.	6	Applying
	Lab Exam – II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	Pye.R.G.W., "Injection Mould Design", Affiliated East – west press pvt Ltd, 2000
R2	Briston and Gosselin, —Introduction to Plastics", Newnes-Butterworths, London, 1970
R3	Mills.N.J., —Plastics", ELBS, 1986
R4	Dominick V.Rosato and Donald V.Rosato., "Injection Moulding Hand Book", CBS Publishers & Distributors, Delhi, 1987 4
R5	Athalye.A.S., "Moulding of Plastics", Multi Tech Publishing Co., 1998

Online Resources:

Sl.No	Website Link
1	http://164.100.133.129:81/econtent/Uploads/13%20Design%20for%20Injection%20Moulding.pdf
2	https://web.itu.edu.tr/gulmezt/imalat%20makinalari/2-heat%20treatment%20II.pdf
3	http://uhv.cheme.cmu.edu/procedures/machining/ch2.pdf
4	https://www.researchgate.net/publication/281442356_HEAT_TREATMENT